

Motor Datasheet: tm SAR

20260127 165414 05AG 20C

123mNm

Operating Point: Continuous | Temperature: 20 °C

Geometry (active elements only)

Parameter	Value	Unit
Machine Outer Diameter	55.0	mm
Machine Inner Diameter	20.0	mm
Machine Thickness / Width	23.7	mm
Air Gap	0.50	mm
Fill Factor (input)	0.65	-
Slot Fill Factor (calculated)	0.58	-
Slot Width (tooth to tooth)	4.292	mm
Available Coil Space	2.146	mm
Assumed Coil Width (simulation)	1.946	mm

Mass Properties (active elements only)

Parameter	Value	Unit
Machine Mass	271.6	g
Rotor Mass	60.3	g
Stator Mass	211.2	g
Rotor Structure Included	No	-
Rotor Moment of Inertia (active elements only)	245.8	g·cm²

Electrical Properties

Parameter	Value	Unit
Supply Current (RMS)	4.024	A
Supply Current (Peak)	5.691	A
Id (Peak)	-0.000	A
Iq (Peak)	5.691	A
Phase Advance Angle	0.0	°elec
Phase Resistance	69.729	mOhm
D-Axis Inductance (Ld)	0.000	μH
Q-Axis Inductance (Lq)	0.000	μH
Terminal Inductance (Ltt = Ld + Lq)	0.000	μH
PM Flux Linkage	0.000	mWb
Frequency at operating point (electrical)	237.500	Hz
Winding Configuration	wye (star)	-
DC Supply Voltage	9.000	V
Power Factor	0.935	-

Performance

Parameter	Value	Unit
Torque (Continuous)	123.2	mNm
Speed	2850.0	rpm
BackEMF Constant (Ke, phase peak)	1.5	mV/rpm
Motor Constant (Km, only resistive losses)	66.9	mNm/√W
Specific Motor Constant (Km / mass)	246.4	mNm/√W/kg
Torque Constant (Kt, peak, no-load)	21.8	mNm/A
Torque Ripple	2.9	%

Saturation	0.8	%
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Losses

Parameter	Value	Unit
Total Losses	4.1	W
Resistive Losses	3.4	W
Iron Core Losses	0.7	W
Magnet Eddy Current Losses	0.0	W
Rotor Structure Eddy Current Losses	0.0	W
Winding Eddy Current Losses	0.0	W

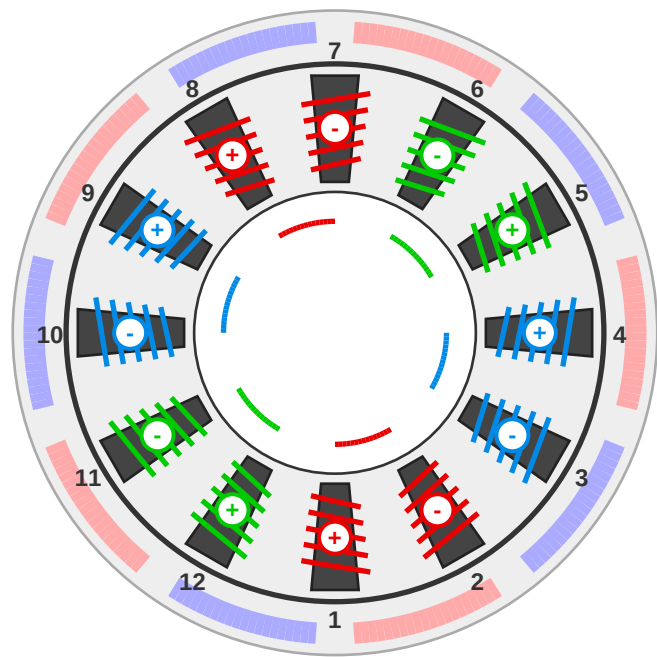
Configuration

Parameter	Value	Unit
Number of Stators	1	-
Number of Rotors	1	-
Number of Slots	12	-
Pole Pairs	5	-
Number of Parallel Paths	1	-
Number of Turns per Tooth (designed incl. parallel paths)	20.0	-
Number of Phase Turns (all phases in series)	80.0	-

Winding Configuration

Schema: **AabBCcaABbcC**

Concentrated Winding | Balanced | Cogging: 60/rev | Ripple: 60/rev



Materials

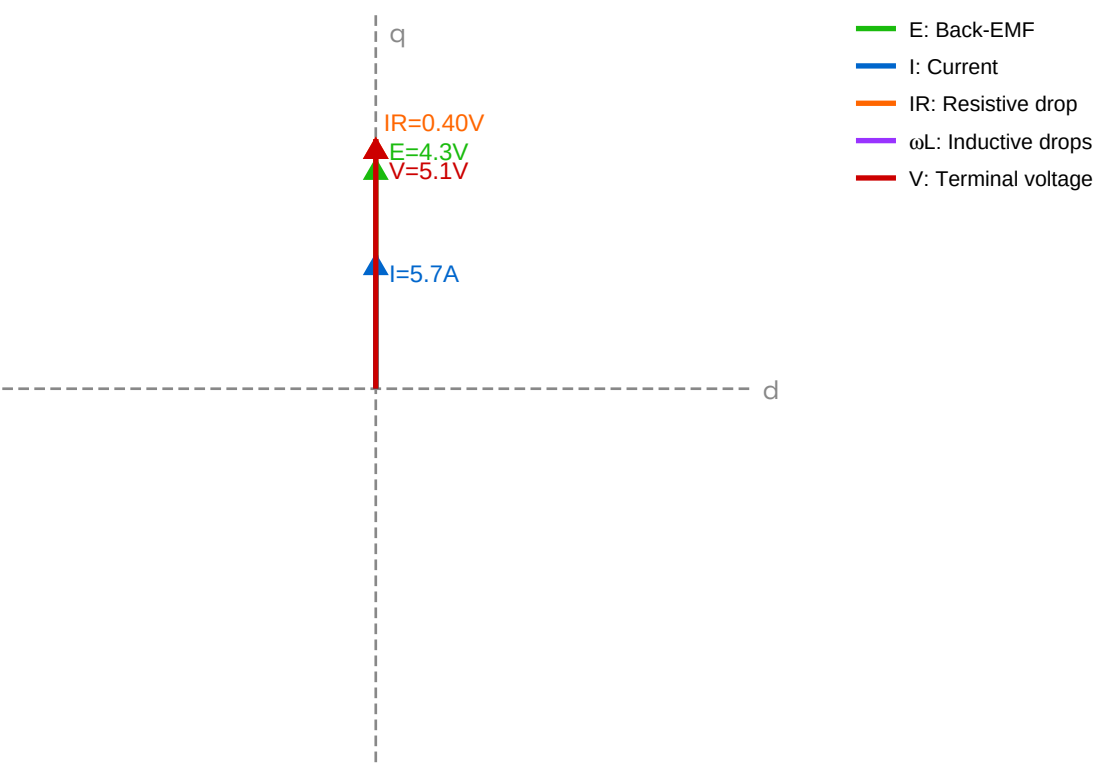
Component	Material	
Stator Core	700HR 5P 800MPa-Hoganas	-
Magnets	BMHFa-32/27	-
Rotor Backiron	430-FR 15mm-Carpenter	-
Coil	Copper	-

Voltage Breakdown at Operating Point (with integer turns)

Parameter	Value	Unit
Back-EMF Voltage (Phase, Peak)	4.341	V
Resistive Drop (Phase, Peak)	0.397	V
Inductive Drop Ld (Peak)	0.000	V
Inductive Drop Lq (Peak)	0.000	V

Inductive Drop Total (Peak)	0.000	V
Terminal Voltage (Phase, Peak)	5.065	V
Back-EMF Voltage (Line-to-Line, Peak)	7.519	V
Resistive Drop (Line-to-Line, Peak)	0.687	V
Inductive Drop (Line-to-Line, Peak)	0.000	V
Terminal Voltage (Line-to-Line, Peak)	8.773	V

Phasor Diagram Phase Values NOT Line-to-Line (d-q reference frame)



Mechanical Properties

Parameter	Value	Unit
Airgap Axial Force (Average)	38.64	N
Magnet Axial Force Positive Maximum	0.00	N
Magnet Axial Force Positive Minimum	0.00	N
Magnet Axial Force Negative Maximum	-14.80	N
Magnet Axial Force Negative Minimum	-11.04	N

Glossary

- Peak — Refers to 0-to-peak amplitude (NOT peak-to-peak). For sinusoidal waveforms: $\text{Peak} = \text{RMS} \times \sqrt{2}$.
- RMS — Root Mean Square, the equivalent DC value for power calculations.
- Line-to-Line (L-L) — Voltage measured between two phases. For wye connection: $V_{LL} = V_{\text{phase}} \times \sqrt{3}$.